Target validation paths — aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2-rerun2

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

Two results decide whether the question this case was submitted to answer has an answer at all. Allele-level HLA-A*02:01 typing gates the Fred Hutch HA-1 TCR-T on NCT03326921, TSC-100/TSC-101 on NCT05473910 and BSB-1001 on NCT06704152, and HA-1/HA-2 genotyping of the patient together with her HLA-identical sibling then decides whether the mismatch those products need actually runs the right way. If both alleles, A*02:01 and the A*24:02 fallback, type negative, this report has no within-scope HLA-restricted workup left to offer, and the next conversation about standard salvage care belongs to the treating team.

HLA-restricted immunotherapy

High-resolution class I typing of HLA-A, -B and -C is essential and it is mostly a records problem. The typing was run for the 2020 sibling transplant and the allele-level call is sitting in a transplant or donor-registry file rather than in the chart on hand, so ask for that report before drawing anything; a fresh NGS type takes 3 to 7 days with STAT available if it cannot be produced. HA-1/HA-2 genotyping (HMHA1 rs1801284 and MYO1G) is the second essential test and it is the slow one. NCT03326921 wants a recipient carrying rs1801284 A/G or A/A with a G/G donor, NCT06704152 wants HA-1 H/H or H/R with an HA-1-negative donor, and HLA-identical siblings routinely differ at these minor loci, so a patient-only result settles nothing. It runs 1 to 3 weeks as a specialised send-out, needs a fresh donor sample and donor consent, and routing it through the Fred Hutch or ALLOHA screening pathway is often faster than a catalogue order.

One resistance question belongs here too, and a late relapse six years after a matched-sibling graft is exactly the history that raises it. HLA loss of heterozygosity on sorted blasts, 6p LOH by NGS or SNP array paired with a flow read of surface class I, asks whether the relapsed clone still presents the restricting allele. If that clone has deleted the haplotype carrying HLA-A*02:01, every A*02:01-restricted product is aimed at something the leukaemia no longer displays, and the germline typing will still read positive. Reflex it once the typing returns. Two further reads confirm targets without gating anything: standardised WT1 RQ-PCR reported as copies per 10^4 ABL, which also sets an MRD baseline that compares across centres, and a PRAME expression read on blasts, which is worth ordering only after A*02:01 is known to be present.

TP53 Y220C

The record holds a pathogenic TP53 mutation from 2019 and a negative relapse panel from 2026, and the workup is built to keep both on the table rather than pick the more recent one. Retrieving the 2019 variant table with its HGVS notation and VAF is essential and costs nothing but a phone call, because NCT06616636 screens on a Y220C call at VAF above 2 percent and the specific amino-acid change is what opens or closes that trial. The second essential step is a copy-number-capable myeloid panel at relapse reporting TP53 at the variant level: the 2026 assay ran at a 2 percent limit of detection with no copy-number calling, so a subclonal or copy-loss event would have been reported exactly as it was. Two orthogonal reads corroborate the call off archival material at no new procedure. 17p/TP53 FISH looks for the copy loss a sequence-only panel cannot see, which matters in a leukaemia whose 2019 karyotype carried der(17)t(13;17). p53 IHC scored by pattern, overexpression versus complete null versus wild-type, adds a protein-level read, though it never names the variant.

CD123

Quantitative CD123 density by flow, reported as antibodies bound per cell or MESF against calibrated beads, is essential and it is a request rather than a default readout. Density across AML spans roughly 200 to 10,000 molecules per cell, so the qualitative positive call on file cannot separate a strong handle from a marginal one, and neither tagraxofusp nor pivekimab sunirine carries an AML label, which puts the weight of the argument on the number. Honest caveat: no validated cutoff predicts benefit from either agent, so the result informs judgement rather than clearing a published threshold.

The right-thigh myeloid sarcoma deserves its own tissue. In a meta-analysis of 85 studies, 56 percent of the 112 patients with paired myeloid sarcoma and marrow sequencing were discordant in mutational profile, and CD33 was detectable in 79 percent of myeloid sarcoma tissue rather than all of it. A core biopsy sent for CD33 and CD123 flow plus targeted myeloid NGS tests whether the marrow speaks for the lesion. The comparison is what carries the information, so sequence it against the marrow work rather than instead of it.

CD33

Quantitative CD33 density belongs on the same specimen and the same requisition as the CD123 request. The central flow reanalysis of ALFA-0701 tied gemtuzumab benefit to the level of blast CD33 expression rather than to positivity, so a low-density case that still reads positive is a weak handle for a CD33 antibody-drug conjugate. Reported against the laboratory's own AML range, not a published cutoff. The rs12459419 splicing-SNP genotype sits at medium priority and reads as context: the T allele shifts splicing toward an isoform missing the IgV domain gemtuzumab binds, the randomised evidence for that is paediatric, and adult replication is inconsistent enough that a T allele should temper a conversation rather than end one.

Second-transplant platform

Three reads support the transplant plan and none of them gates enrolment. STR chimerism on flow-sorted CD34+ blasts confirms the relapse is recipient-origin rather than donor-derived, and the sorting is not optional detail: a sex-matched sibling donor rules out XY FISH, and unsorted chimerism gets diluted by normal donor haematopoiesis. A full metaphase karyotype with MECOM and KMT2A FISH puts the 2026 findings into a current cytogenetic picture that trial screens read directly. The germline hereditary myeloid malignancy panel is the one worth starting early, because it bears on re-using the same sibling. A predisposition variant shared with the donor risks donor-derived disease and would redirect donor choice, a germline TP53 finding would change conditioning intensity, and the specimen requirement is the catch: with blasts at 58 to 82 percent, blood and saliva cannot give a clean germline call, so the panel wants cultured skin fibroblasts from a punch biopsy and culture adds 4 to 6 weeks.

Biomarkers measured, but not to decision resolution

Three markers have something on file that cannot carry a decision as it stands, and each is a different kind of short.

- HLA class I genotype. Typing was performed for the 2020 transplant, so the result exists somewhere; what is missing is the allele-level call in a form anyone can act on. The fix is retrieval first, high-resolution NGS typing of both parties second, and the same pass covers the sibling because HA-1/HA-2 eligibility depends on the donor as much as the patient.
- CD123. Flow called the blasts positive and stopped there. The immunotoxin and ADC literature ties benefit to antigen density, not to a positive-or-negative call, so quantitative ABC or MESF on a current marrow or blood specimen, with the percentage of blasts positive stated alongside it, is what closes the gap.

- PRAME. This one is not under-resolved, it is unmeasured, and two gaps stack. The restricting allele is unknown and expression on these blasts has never been assessed. Order the class I typing first; if A*02:01 is present, reflex PRAME by RT-qPCR on sorted blasts or IHC on the marrow clot section. Reported overexpression in AML swings widely with assay and cutoff, so quantify against the laboratory's own reference rather than assuming the target is there.

Where to order these assays

Assay	Provider	Decision gated	Contact
High-resolution (4-digit) HLA class I typing of HLA-A, -B and -C in the patient and the HLA-identical sibling donor	**HistoGenetics (preferred) (HLA sequence-based typing (NGS))**	**TSC-100/TSC-101 (NCT05473910), Fred Hutch HA-1 TCR-T (NCT03326921) and BSB-1001 (NCT06704152), all of which require HLA-A*02:01**	**test info · 300 Executive Blvd, Ossining, NY 10562 · (914) 762-0300**
High-resolution (4-digit) HLA class I typing of HLA-A, -B and -C in the patient and the HLA-identical sibling donor	Versiti Diagnostic Laboratories (High-resolution HLA typing by NGS)	TSC-100/TSC-101 (NCT05473910), Fred Hutch HA-1 TCR-T (NCT03326921) and BSB-1001 (NCT06704152), all of which require HLA-A*02:01	test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117
High-resolution (4-digit) HLA class I typing of HLA-A, -B and -C in the patient and the HLA-identical sibling donor	Labcorp	TSC-100/TSC-101 (NCT05473910), Fred Hutch HA-1 TCR-T (NCT03326921) and BSB-1001 (NCT06704152), all of which require HLA-A*02:01	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
High-resolution (4-digit) HLA class I typing of HLA-A, -B and -C in the patient and the HLA-identical sibling donor	ARUP Laboratories	TSC-100/TSC-101 (NCT05473910), Fred Hutch HA-1 TCR-T (NCT03326921) and BSB-1001 (NCT06704152), all of which require HLA-A*02:01	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · (800) 522-2787
HA-1 / HA-2 minor histocompatibility antigen genotyping (HMHA1 rs1801284 and MYO1G) in the patient and the sibling donor	**Fred Hutchinson Cancer Center (Immunogenetics / Bleakley HA-1 program) (preferred)**	**TSC-100/TSC-101 (NCT05473910), Fred Hutch HA-1 TCR-T (NCT03326921) and BSB-1001 (NCT06704152); all require recipient HA-1-positive with an HA-1-negative donor**	**test info · 1100 Fairview Ave N, Seattle, WA 98109 · (206) 667-5000**
HA-1 / HA-2 minor histocompatibility antigen genotyping (HMHA1 rs1801284 and MYO1G) in the patient and the sibling donor	TScan Therapeutics (ALLOHA trial screening)	TSC-100/TSC-101 (NCT05473910), Fred Hutch HA-1 TCR-T (NCT03326921) and BSB-1001 (NCT06704152); all require recipient HA-1-positive with an HA-1-negative donor	test info · 830 Winter St, Waltham, MA 02451
HA-1 / HA-2 minor histocompatibility antigen genotyping (HMHA1 rs1801284 and MYO1G) in the patient and the sibling donor	Versiti Diagnostic Laboratories	TSC-100/TSC-101 (NCT05473910), Fred Hutch HA-1 TCR-T (NCT03326921) and BSB-1001 (NCT06704152); all require recipient HA-1-positive with an HA-1-negative donor	test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117

HA-1 / HA-2 minor histocompatibility antigen genotyping (HMHA1 rs1801284 and MYO1G) in the patient and the sibling donor	HistoGenetics (HLA sequence-based typing (NGS))	TSC-100/TSC-101 (NCT05473910), Fred Hutch HA-1 TCR-T (NCT03326921) and BSB-1001 (NCT06704152); all require recipient HA-1-positive with an HA-1-negative donor	test info · 300 Executive Blvd, Ossining, NY 10562 · (914) 762-0300
Retrieval of the 2019 diagnostic TP53 variant identity and VAF from the original NGS report	Not a send-out. Medical-records and laboratory-report request to whichever laboratory ran the 2019 diagnostic panel	Rezatapopt plus azacitidine via NCT06616636, which is Y220C-contingent	Request the annotated variant table (HGVS p. notation and VAF) from the 2019 testing laboratory
Copy-number-capable myeloid NGS panel at relapse with TP53 variant-level reporting and VAF (Y220C determination)	**Foundation Medicine (preferred) (FoundationOne Heme)**	**Rezatapopt plus azacitidine via NCT06616636; also settles the TP53 adverse-risk assignment for HCT2 planning**	**test info · 150 Second St, Cambridge, MA 02141 · (888) 988-3639**
Copy-number-capable myeloid NGS panel at relapse with TP53 variant-level reporting and VAF (Y220C determination)	Tempus (xT / xF)	Rezatapopt plus azacitidine via NCT06616636; also settles the TP53 adverse-risk assignment for HCT2 planning	test info · 600 W Chicago Ave, Suite 510, Chicago, IL 60654 · (800) 739-4137
Copy-number-capable myeloid NGS panel at relapse with TP53 variant-level reporting and VAF (Y220C determination)	NeoGenomics Laboratories (NEO Comprehensive Myeloid)	Rezatapopt plus azacitidine via NCT06616636; also settles the TP53 adverse-risk assignment for HCT2 planning	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
Copy-number-capable myeloid NGS panel at relapse with TP53 variant-level reporting and VAF (Y220C determination)	Labcorp	Rezatapopt plus azacitidine via NCT06616636; also settles the TP53 adverse-risk assignment for HCT2 planning	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
Copy-number-capable myeloid NGS panel at relapse with TP53 variant-level reporting and VAF (Y220C determination)	Mayo Clinic Laboratories	Rezatapopt plus azacitidine via NCT06616636; also settles the TP53 adverse-risk assignment for HCT2 planning	test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710
Quantitative CD123 (IL3RA) antigen-density flow cytometry reported as antibodies bound per cell (ABC) or MESF	**Hematologics, Inc. (preferred) (Differences from Normal (DfN) flow)**	**Tagraxofusp and pivekimab sunirine (CD123-directed), and dual CD33/CD123 strategies**	**test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · (800) 860-0934**
Quantitative CD123 (IL3RA) antigen-density flow cytometry reported as antibodies bound per cell (ABC) or MESF	NeoGenomics Laboratories	Tagraxofusp and pivekimab sunirine (CD123-directed), and dual CD33/CD123 strategies	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
Quantitative CD123 (IL3RA) antigen-density flow cytometry reported as antibodies bound per cell (ABC) or MESF	Labcorp	Tagraxofusp and pivekimab sunirine (CD123-directed), and dual CD33/CD123 strategies	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
Quantitative CD123 (IL3RA) antigen-density flow cytometry reported as antibodies bound per cell (ABC) or MESF	Mayo Clinic Laboratories	Tagraxofusp and pivekimab sunirine (CD123-directed), and dual CD33/CD123 strategies	test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710
Quantitative CD33 antigen-density flow cytometry reported as antibodies bound per cell (ABC) or MESF	**Hematologics, Inc. (preferred) (Differences from Normal (DfN) flow)**	**Gemtuzumab ozogamicin (CD33 ADC) and dual CD33/CD123 targeting**	**test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · (800) 860-0934**

Quantitative CD33 antigen-density flow cytometry reported as antibodies bound per cell (ABC) or MESF	NeoGenomics Laboratories	Gemtuzumab ozogamicin (CD33 ADC) and dual CD33/CD123 targeting	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
Quantitative CD33 antigen-density flow cytometry reported as antibodies bound per cell (ABC) or MESF	Labcorp	Gemtuzumab ozogamicin (CD33 ADC) and dual CD33/CD123 targeting	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
Quantitative CD33 antigen-density flow cytometry reported as antibodies bound per cell (ABC) or MESF	Mayo Clinic Laboratories	Gemtuzumab ozogamicin (CD33 ADC) and dual CD33/CD123 targeting	test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710
Quantitative WT1 transcript by the ELN-standardised RQ-PCR assay (WT1 copies per 10^4 ABL)	**ARUP Laboratories (preferred)**	**WT1-directed T-cell approaches; also sets the standardised MRD baseline for post-HCT2 monitoring**	**test info · 500 Chipeta Way, Salt Lake City, UT 84108 · (800) 522-2787**
Quantitative WT1 transcript by the ELN-standardised RQ-PCR assay (WT1 copies per 10^4 ABL)	MD Anderson Cancer Center, Molecular Diagnostics Laboratory	WT1-directed T-cell approaches; also sets the standardised MRD baseline for post-HCT2 monitoring	test info · 1515 Holcombe Blvd, Houston, TX 77030 · (877) 632-6789
Quantitative WT1 transcript by the ELN-standardised RQ-PCR assay (WT1 copies per 10^4 ABL)	Mayo Clinic Laboratories	WT1-directed T-cell approaches; also sets the standardised MRD baseline for post-HCT2 monitoring	test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710
Quantitative WT1 transcript by the ELN-standardised RQ-PCR assay (WT1 copies per 10^4 ABL)	NeoGenomics Laboratories	WT1-directed T-cell approaches; also sets the standardised MRD baseline for post-HCT2 monitoring	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
PRAME expression on blasts by quantitative RT-PCR or RNA sequencing (reflex after HLA class I typing)	**NeoGenomics Laboratories (preferred) (PRAME (IHC / RNA))**	**PRAME-directed TCR-T or ImmTAC candidacy; reflex only if HLA-A*02:01 types positive**	**test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907**
PRAME expression on blasts by quantitative RT-PCR or RNA sequencing (reflex after HLA class I typing)	Tempus (xR RNA)	PRAME-directed TCR-T or ImmTAC candidacy; reflex only if HLA-A*02:01 types positive	test info · 600 W Chicago Ave, Suite 510, Chicago, IL 60654 · (800) 739-4137
PRAME expression on blasts by quantitative RT-PCR or RNA sequencing (reflex after HLA class I typing)	MD Anderson Cancer Center, Molecular Diagnostics Laboratory	PRAME-directed TCR-T or ImmTAC candidacy; reflex only if HLA-A*02:01 types positive	test info · 1515 Holcombe Blvd, Houston, TX 77030 · (877) 632-6789
PRAME expression on blasts by quantitative RT-PCR or RNA sequencing (reflex after HLA class I typing)	Mayo Clinic Laboratories	PRAME-directed TCR-T or ImmTAC candidacy; reflex only if HLA-A*02:01 types positive	test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710
Biopsy of the right-thigh myeloid sarcoma with CD33 and CD123 immunophenotyping plus targeted myeloid NGS	**NeoGenomics Laboratories (preferred)**	**Whether CD33/CD123-directed and TP53-directed choices made on marrow also cover the extramedullary lesion**	**test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907**

Biopsy of the right-thigh myeloid sarcoma with CD33 and CD123 immunophenotyping plus targeted myeloid NGS	Labcorp	Whether CD33/CD123-directed and TP53-directed choices made on marrow also cover the extramedullary lesion	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
Biopsy of the right-thigh myeloid sarcoma with CD33 and CD123 immunophenotyping plus targeted myeloid NGS	Mayo Clinic Laboratories	Whether CD33/CD123-directed and TP53-directed choices made on marrow also cover the extramedullary lesion	test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710
Biopsy of the right-thigh myeloid sarcoma with CD33 and CD123 immunophenotyping plus targeted myeloid NGS	Quest Diagnostics	Whether CD33/CD123-directed and TP53-directed choices made on marrow also cover the extramedullary lesion	test info · 500 Plaza Dr, Secaucus, NJ 07094 · (866) 697-8378
17p / TP53 deletion FISH at relapse	**Mayo Clinic Laboratories (preferred)**	**TP53 copy loss and multi-hit allelic state; supports the TP53-aberrant call and the adverse-risk assignment**	**test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710**
17p / TP53 deletion FISH at relapse	Labcorp	TP53 copy loss and multi-hit allelic state; supports the TP53-aberrant call and the adverse-risk assignment	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
17p / TP53 deletion FISH at relapse	NeoGenomics Laboratories	TP53 copy loss and multi-hit allelic state; supports the TP53-aberrant call and the adverse-risk assignment	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
17p / TP53 deletion FISH at relapse	Quest Diagnostics	TP53 copy loss and multi-hit allelic state; supports the TP53-aberrant call and the adverse-risk assignment	test info · 500 Plaza Dr, Secaucus, NJ 07094 · (866) 697-8378
p53 protein immunohistochemistry on the marrow biopsy, scored by pattern (overexpression versus complete null)	**Mayo Clinic Laboratories (preferred)**	**Corroborates the TP53-aberrant call feeding the rezatapopt decision; not variant-level**	**test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710**
p53 protein immunohistochemistry on the marrow biopsy, scored by pattern (overexpression versus complete null)	NeoGenomics Laboratories	Corroborates the TP53-aberrant call feeding the rezatapopt decision; not variant-level	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
p53 protein immunohistochemistry on the marrow biopsy, scored by pattern (overexpression versus complete null)	Labcorp	Corroborates the TP53-aberrant call feeding the rezatapopt decision; not variant-level	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
p53 protein immunohistochemistry on the marrow biopsy, scored by pattern (overexpression versus complete null)	Quest Diagnostics	Corroborates the TP53-aberrant call feeding the rezatapopt decision; not variant-level	test info · 500 Plaza Dr, Secaucus, NJ 07094 · (866) 697-8378

STR chimerism on flow-sorted CD34+ blasts (recipient versus donor origin)	**Versiti Diagnostic Laboratories (preferred)**	**Clonal origin of the relapse; informs HCT2 versus DLI versus alternative-donor strategy**	**test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117**
STR chimerism on flow-sorted CD34+ blasts (recipient versus donor origin)	CareDx (AlloSeq HCT)	Clonal origin of the relapse; informs HCT2 versus DLI versus alternative-donor strategy	test info · 8000 Marina Blvd, Suite 100, Brisbane, CA 94005 · (415) 287-2300
STR chimerism on flow-sorted CD34+ blasts (recipient versus donor origin)	Mayo Clinic Laboratories	Clonal origin of the relapse; informs HCT2 versus DLI versus alternative-donor strategy	test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710
STR chimerism on flow-sorted CD34+ blasts (recipient versus donor origin)	Labcorp	Clonal origin of the relapse; informs HCT2 versus DLI versus alternative-donor strategy	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
HLA loss-of-heterozygosity assessment on sorted blasts (6p LOH by NGS or SNP array) with HLA class I surface expression by flow	**Versiti Diagnostic Laboratories (preferred) (HLA loss of heterozygosity evaluation)**	**Whether HLA-A*02:01-restricted TCR-T stays on-target in the relapsed clone; also informs DLI and donor choice for HCT2**	**test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117**
HLA loss-of-heterozygosity assessment on sorted blasts (6p LOH by NGS or SNP array) with HLA class I surface expression by flow	CareDx (AlloSeq HCT)	Whether HLA-A*02:01-restricted TCR-T stays on-target in the relapsed clone; also informs DLI and donor choice for HCT2	test info · 8000 Marina Blvd, Suite 100, Brisbane, CA 94005 · (415) 287-2300
HLA loss-of-heterozygosity assessment on sorted blasts (6p LOH by NGS or SNP array) with HLA class I surface expression by flow	Fred Hutchinson Cancer Center (Immunogenetics / Bleakley HA-1 program)	Whether HLA-A*02:01-restricted TCR-T stays on-target in the relapsed clone; also informs DLI and donor choice for HCT2	test info · 1100 Fairview Ave N, Seattle, WA 98109 · (206) 667-5000
HLA loss-of-heterozygosity assessment on sorted blasts (6p LOH by NGS or SNP array) with HLA class I surface expression by flow	NeoGenomics Laboratories	Whether HLA-A*02:01-restricted TCR-T stays on-target in the relapsed clone; also informs DLI and donor choice for HCT2	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
Full metaphase karyotype at relapse with MECOM and KMT2A FISH	**Mayo Clinic Laboratories (preferred)**	**Adverse-cytogenetics confirmation (MECOM, KMT2A amplification, complex karyotype) for prognosis and trial stratification**	**test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710**
Full metaphase karyotype at relapse with MECOM and KMT2A FISH	NeoGenomics Laboratories	Adverse-cytogenetics confirmation (MECOM, KMT2A amplification, complex karyotype) for prognosis and trial stratification	test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907
Full metaphase karyotype at relapse with MECOM and KMT2A FISH	Labcorp	Adverse-cytogenetics confirmation (MECOM, KMT2A amplification, complex karyotype) for prognosis and trial stratification	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167

Full metaphase karyotype at relapse with MECOM and KMT2A FISH	Quest Diagnostics	Adverse-cytogenetics confirmation (MECOM, KMT2A amplification, complex karyotype) for prognosis and trial stratification	test info · 500 Plaza Dr, Secaucus, NJ 07094 · (866) 697-8378
Germline hereditary myeloid malignancy panel (TP53, DDX41, RUNX1, CEBPA, GATA2, ANKRD26, ETV6 and related genes) on cultured skin fibroblasts	**University of Chicago Genetic Services Laboratories (preferred) (Hereditary Myeloid Malignancy Panel)**	**Sibling-donor selection and conditioning intensity for HCT2; family screening**	**test info · 5841 S Maryland Ave, Room G701, MC 0077, Chicago, IL 60637 · (773) 834-0555**
Germline hereditary myeloid malignancy panel (TP53, DDX41, RUNX1, CEBPA, GATA2, ANKRD26, ETV6 and related genes) on cultured skin fibroblasts	GeneDx	Sibling-donor selection and conditioning intensity for HCT2; family screening	test info · 207 Perry Pkwy, Gaithersburg, MD 20877 · (888) 729-1206
Germline hereditary myeloid malignancy panel (TP53, DDX41, RUNX1, CEBPA, GATA2, ANKRD26, ETV6 and related genes) on cultured skin fibroblasts	Labcorp (Labcorp Genetics (Invitae) hereditary haematologic panel)	Sibling-donor selection and conditioning intensity for HCT2; family screening	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
Germline hereditary myeloid malignancy panel (TP53, DDX41, RUNX1, CEBPA, GATA2, ANKRD26, ETV6 and related genes) on cultured skin fibroblasts	Mayo Clinic Laboratories	Sibling-donor selection and conditioning intensity for HCT2; family screening	test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710
CD33 rs12459419 splicing-SNP genotype	**Hematologics, Inc. (preferred) (CD33 Single Nucleotide Polymorphism Analysis for Gemtuzumab Ozogamicin Treatment Response)**	**Refines the expected gemtuzumab ozogamicin benefit; does not gate enrolment or prescribing**	**test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · (800) 860-0934**
CD33 rs12459419 splicing-SNP genotype	Labcorp	Refines the expected gemtuzumab ozogamicin benefit; does not gate enrolment or prescribing	test info · 358 South Main St, Burlington, NC 27215 · (800) 845-6167
CD33 rs12459419 splicing-SNP genotype	ARUP Laboratories	Refines the expected gemtuzumab ozogamicin benefit; does not gate enrolment or prescribing	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · (800) 522-2787

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements

High-resolution (4-digit) HLA class I typing of HLA-A, -B and -C in the patient and the HLA-identical sibling donor	Resolves the 4-digit HLA-A02:01 and A24:02 status that every HLA-restricted option named for this case sits behind: TSC-100/TSC-101 (NCT05473910), the Fred Hutch HA-1 TCR-T (NCT03326921) and BSB-1001 (NCT06704152) each require HLA-A02:01, and A24:02 keeps a WT1-siTCR route open if A*02:01 is negative. High-resolution typing was performed for the 2020 sibling transplant but sits in the record as a placeholder, so pull the allele-level report from the transplant and donor-registry file first and re-type on a fresh draw only if it cannot be produced quickly. Type the sibling donor in the same pass; until an allele call exists the question this case was submitted to answer cannot be scoped in either direction.	HistoGenetics (HLA sequence-based typing (NGS)) · test info · 300 Executive Blvd, Ossining, NY 10562 · (914) 762-0300	5-10 mL whole blood (EDTA) from the patient and from the sibling donor; buccal swab acceptable for the donor

HA-1 / HA-2 minor histocompatibility antigen genotyping (HMHA1 rs1801284 and MYO1G) in the patient and the sibling donor	Establishes the recipient-positive / donor-negative disparity that HA-directed TCR-T requires. NCT03326921 enrolls recipients carrying the HA-1(H) genotype (rs1801284 A/G or A/A) whose donor is HA-1(H) negative (G/G); NCT05473910 requires an HA-1-positive genotype for TSC-100 and an HA-2-positive genotype for TSC-101; NCT06704152 requires HA-1 H/H or H/R with an HA-1-negative donor. HLA-identical siblings can still differ at these minor loci, so a patient-only result leaves eligibility undetermined, and the donor sample and consent are the slow step worth starting now.	Fred Hutchinson Cancer Center (Immunogenetics / Bleakley HA-1 program) · test info · 1100 Fairview Ave N, Seattle, WA 98109 · (206) 667-5000	5-10 mL whole blood (EDTA) from the patient and from the sibling donor; buccal swab acceptable for the donor
Retrieval of the 2019 diagnostic TP53 variant identity and VAF from the original NGS report	Recovers the exact amino-acid change and VAF from the 2019 diagnostic NGS, which is the cheapest route to whether the founder TP53 lesion was Y220C. NCT06616636 (rezatapopt plus azacitidine in TP53 Y220C myeloid malignancies) requires a Y220C call by CLIA-approved local testing at VAF above 2 percent, so the specific variant rather than TP53 status in general is what opens or closes that door. This is a records request against an existing report, and it should happen before any conclusion is drawn from the 2026 negative result.	2019 testing laboratory (records request); ask for the annotated variant table with HGVS p. notation and VAF	No new specimen. Request the annotated variant table (HGVS p. notation and VAF) from the 2019 testing laboratory

myeloid NGS panel at relapse with TP53 variant-level reporting and VAF (Y220C determination)	Re-tests TP53 at relapse on an assay that can see what the 2026 panel could not: that panel ran at a 2 percent limit of detection with no copy-number calling and was not MRD-grade, so a subclonal or copy-loss TP53 event would have been reported as negative. A copy-number-capable panel reporting variant identity and VAF resolves the 2019-mutant versus 2026-negative discrepancy in whichever direction the biology actually runs, and a Y220C call above 2 percent VAF from a CLIA laboratory is the entry criterion for NCT06616636. Down-calling to TP53 wild-type on the 2026 result alone would discard both a possible rezatapopt handle and the adverse-risk assignment that shapes the transplant plan.	Foundation Medicine (FoundationOne Heme) · test info · 150 Second St, Cambridge, MA 02141 · (888) 988-3639	Fresh bone marrow aspirate at relapse (EDTA); peripheral blood acceptable given circulating blasts of 58-82 percent

Quantitative CD123 (IL3RA) antigen-density flow cytometry reported as antibodies bound per cell (ABC) or MESF	Turns the qualitative CD123-positive flow call into the surface-density number that CD123-directed agents are actually selected on. Reported CD123 density across AML spans roughly 200 to 10,000 molecules per cell, so a yes-or-no call cannot separate a strong handle from a marginal one, and both tagraxofusp and pivekimab sunirine carry BPDCN rather than AML labels, which puts the weight of any off-label or trial argument on the density read. Request ABC or MESF against calibrated beads explicitly, with the percentage of blasts positive alongside it; density is not a default readout on a phenotyping panel.	Hematologics, Inc. (Differences from Normal (DfN) flow) · test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · (800) 860-0934	Fresh bone marrow aspirate or peripheral blood (sodium heparin or EDTA), viable cells, shipped same day
Quantitative CD33 antigen-density flow cytometry reported as antibodies bound per cell (ABC) or MESF	Turns the qualitative CD33-positive call into a density measure. In the central flow reanalysis of the randomised ALFA-0701 trial, benefit from gemtuzumab ozogamicin tracked with the level of blast CD33 expression rather than with positivity alone, so a low-density case that still reads positive is a weak handle for a CD33 antibody-drug conjugate. Order it on the same specimen as the CD123 density request; without it a gemtuzumab decision rests on an assay never sized to make it.	Hematologics, Inc. (Differences from Normal (DfN) flow) · test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · (800) 860-0934	Fresh bone marrow aspirate or peripheral blood (sodium heparin or EDTA), viable cells; same specimen as the CD123 density request

Quantitative WT1 transcript by the ELN-standardised RQ-PCR assay (WT1 copies per 10^4 ABL)	Confirms that the blasts overexpress WT1, the antigen behind the WT1-directed approaches this case names, and sets a quantitative baseline for tracking disease after any consolidation. The ELN-standardised assay has published reference thresholds of 250 copies per 10^4 ABL in marrow and 50 per 10^4 ABL in blood, so the number compares across centres instead of being laboratory-specific. Note that the WT1-C4 trial named in the profile (NCT01640301) has terminated, so this read currently confirms the target and supports MRD monitoring rather than unlocking a named product.	ARUP Laboratories · test info · 500 Chipeta Way, Salt Lake City, UT 84108 · (800) 522-2787	2-3 mL bone marrow or 5-10 mL peripheral blood in EDTA, delivered within 24-48 hours for RNA integrity

PRAME expression on blasts by quantitative RT-PCR or RNA sequencing (reflex after HLA class I typing)	Answers whether these blasts express PRAME at all, which the record has never measured and which only becomes interpretable once HLA-A02:01 status is known. Reported PRAME overexpression in AML varies widely with assay and cutoff, from roughly a third of cases in some series to 87 percent at first diagnosis in the Steger cohort, where the highest levels sat with unfavourable cytogenetics and with AML arising from myelodysplasia, which is this patient's history. MDG1011, the only PRAME TCR-T taken into myeloid disease, required HLA-A02:01 plus PRAME mRNA detectable by qPCR and its trial (NCT03503968) has terminated, so run this as a reflex on tissue already being sent rather than as a standalone order.	NeoGenomics Laboratories (PRAME (IHC / RNA)) · test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907	Bone marrow aspirate or high-blast peripheral blood (EDTA); FFPE marrow clot section acceptable if IHC is used instead

Biopsy of the right-thigh myeloid sarcoma with CD33 and CD123 plus targeted myeloid NGS	Tests whether the marrow immunophenotype and genotype speak for the extramedullary disease. In a meta-analysis of 85 studies, 56 percent of the 112 patients with paired myeloid sarcoma and marrow sequencing were discordant in mutational profile, CD33 was detectable in 79 percent of myeloid sarcoma tissue rather than all of it, and isolated extramedullary relapse accounted for 46 percent of post-transplant relapses, so an antigen-directed or variant-directed choice made on marrow alone may not cover the thigh lesion. If the biopsy is not feasible, record that limitation in the plan and follow the lesion separately on imaging.	NeoGenomics Laboratories · test info · 12701 Commonwealth Dr, Fort Myers, FL 33913 · (866) 776-5907	Image-guided core biopsy of the right-thigh lesion; one core in RPMI for flow plus FFPE blocks for IHC and NGS
17p / TP53 deletion FISH at relapse	Detects the 17p/TP53 copy loss that a 2 percent LOD sequencing panel without copy-number calling cannot see, which is the most mechanical explanation for a negative relapse result in a leukaemia whose 2019 karyotype carried der(17)t(13;17). It also feeds the allelic-state question, since TP53 mutation plus 17p deletion is a multi-hit state that carries its own risk assignment under the 2022 ELN framework. Leaving it out leaves the leading explanation for the 2019-versus-2026 discrepancy untested.	Mayo Clinic Laboratories · test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710	Bone marrow aspirate (sodium heparin), 2-3 mL; archival FFPE clot or unstained slides acceptable

p53 protein on the marrow biopsy, scored by pattern (overexpression versus complete null)	Adds a protein-level read on material already in the archive, so it costs no new procedure. Pattern-based scoring is what carries the information: strong diffuse nuclear overexpression tracks with missense TP53 mutations and a complete-null pattern with truncating ones, while a normal wild-type pattern argues against a TP53-aberrant state. IHC does not name the variant and its concordance with sequencing is imperfect, so it corroborates rather than substitutes for the variant-level answer the rezatapopt question needs.	Mayo Clinic Laboratories · test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710	Archival FFPE bone marrow core or clot section; no new procedure required
STR chimerism on flow-sorted CD34+ blasts (recipient versus donor origin)	Confirms that the relapse is recipient-origin leukaemia rather than a donor-derived or therapy-related second neoplasm, which the shared del(5q) and del(7q) lesions suggest but do not prove. The sex-matched sibling donor rules out XY FISH as a shortcut, and unsorted chimerism is diluted by normal donor haematopoiesis, so the analysis has to run on sorted CD34+ cells, where declining donor chimerism precedes haematologic relapse by a median of about two months. A donor-derived leukaemia would redirect the plan toward a different donor rather than toward DLI or HA-directed cells from the same sibling.	Versiti Diagnostic Laboratories · test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117	Fresh bone marrow (EDTA) with CD34+ flow sorting, plus the archived donor STR profile from the 2020 transplant

HLA assessment on sorted blasts (6p LOH by NGS or SNP array) with HLA class I surface expression by flow	Asks whether the relapsed clone still presents the restricting allele at all. Genomic HLA loss and transcriptional downregulation of HLA are established routes to relapse after allogeneic transplant, genomic loss having been documented in matched-unrelated as well as mismatched settings, and a blast population that has lost the haplotype carrying HLA-A02:01 is invisible to every A02:01-restricted TCR-T and to donor T cells generally, even when germline typing reads positive. Reflex it on sorted blasts once typing returns; skipping it risks committing to a cellular product whose target the leukaemia has already deleted, which is the pattern a late relapse six years after a matched-sibling graft should raise.	Versiti Diagnostic Laboratories (HLA loss of heterozygosity evaluation) · test info · 638 N 18th St, Milwaukee, WI 53233 · (800) 245-3117	Fresh bone marrow aspirate or high-blast peripheral blood (EDTA) with blast sorting, plus an archived donor or germline reference sample
Full metaphase karyotype at relapse with MECOM and KMT2A FISH	Places the MECOM gain and KMT2A amplification reported in 2026 into a complete cytogenetic picture alongside a current metaphase count. MECOM rearrangement and complex karyotype both carry adverse risk under ELN 2022, and many AML trials read karyotype directly as a stratification or eligibility variable. Without a current full karyotype the risk assignment and several trial screens rest on partial data.	Mayo Clinic Laboratories · test info · 3050 Superior Dr NW, Rochester, MN 55901 · (800) 533-1710	Fresh bone marrow aspirate (sodium heparin), 2-3 mL, delivered within 24 hours for culture

Germline hereditary myeloid malignancy panel (TP53, DDX41, RUNX1, CEBPA, GATA2, ANKRD26, ETV6 and related genes) on cultured skin fibroblasts	Asks whether an inherited predisposition sits under the MDS-to-AML course, which bears directly on a second sibling-donor transplant: a variant shared with the donor risks donor-derived disease and would redirect donor selection, and a germline TP53 finding would change conditioning intensity and trigger family counselling. Specimen source is the load-bearing detail; with blasts at 58 to 82 percent, blood and saliva cannot give a clean germline call, and the University of Chicago laboratory explicitly declines blood or saliva for this panel in patients with a history of MDS or leukaemia and asks for cultured skin fibroblasts. Skipping it leaves the donor question resting on an assumption at exactly the point where it is being re-made.	University of Chicago Genetic Services Laboratories (Hereditary Myeloid Malignancy Panel) · test info · 5841 S Maryland Ave, Room G701, MC 0077, Chicago, IL 60637 · (773) 834-0555	Cultured skin fibroblasts from a 3-4 mm punch biopsy (preferred while blasts circulate); whole blood only if a documented remission sample exists
CD33 rs12459419 splicing-SNP genotype	Genotypes the CD33 splicing SNP rs12459419, where the T allele shifts splicing toward a CD33 isoform lacking the IgV domain that gemtuzumab ozogamicin binds. In the randomised COG AAML0531 trial the CC genotype was associated with benefit from gemtuzumab and T-allele carriers were not, though adult confirmation is mixed and the PETHEMA consolidation analysis found no such association, so this reads as context rather than a gate. It adjusts how much to expect from a CD33 antibody-drug conjugate alongside the density result.	Hematologics, Inc. (CD33 Single Nucleotide Polymorphism Analysis for Gemtuzumab Ozogamicin Treatment Response) · test info · 3161 Elliott Ave, Suite 200, Seattle, WA 98121 · (800) 860-0934	Whole blood or bone marrow; buccal swab acceptable since the result is germline

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.